

Trupti Kularkar

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PROFILE

Data Engineer and **AI/ML Engineer** with 1.6+ years of experience designing and deploying **scalable data pipelines** and **AI-powered automation** on **Google Cloud Platform (GCP)**. Proven track record in **real-time data processing**, **ETL pipeline development**, and **MLOps** using Vertex AI and Gemini API. Strong expertise in **Python**, **SQL**, **BigQuery**, **Apache Airflow**, and **Generative AI** — with a focus on **reducing operational overhead** and improving **data quality** and **system reliability** at scale.

TECHNICAL SKILLS

Language: Python, SQL

Cloud & GCP: BigQuery, Cloud Storage, Pub/Sub, Cloud Functions, Cloud Run, Airflow DAG, Dataflow, ETL Pipelines, Data Warehousing, Real-time Analytics, CI/CD

Generative AI: Vertex AI, Gemini API, Prompt Engineering, AI Agents, LLM, LLM Orchestration, MLOps

ML & AI: TensorFlow, PyTorch, Scikit-Learn, OpenCV, NLP (NLTK), Data Pipeline, Data Quality

Tools: Git, GitHub, Jupyter Notebook, Google Colab, VS Code, Looker Studio

EXPERIENCE

Programmer Analyst, *Cognizant Technology Solutions* 11/2025 – Present | Pune, India

- Built an **LLM-powered ticket resolution agent** on Vertex AI (Gemini 2.5 Pro) that automated ~60% of repetitive support tickets, cutting mean resolution time from 4 hours to under 30 minutes.
- Designed SQL-based data validation frameworks across 15+ ETL pipelines, enforcing row-count reconciliation, null checks, duplicate detection, and schema consistency — reducing data quality incidents by ~35%.
- Optimized BigQuery query costs by ~25% via partitioning and clustering; automated 20+ Airflow DAGs, reducing manual operational overhead by 40% and cutting pipeline failure response time by 50%.
- Architected event-driven, real-time ETL pipelines (Cloud Storage → Pub/Sub → Cloud Functions → BigQuery) processing 500K+ daily events with sub-2-minute end-to-end latency.
- Collaborated with cross-functional teams to integrate AI-based monitoring into enterprise workflow systems.
- Mentored junior engineers on GCP tooling and Airflow pipeline development.

Programmer Analyst Trainee, *Cognizant Technology Solutions* 11/2024 – 11/2025 | Chennai, India

- Gained hands-on experience in GCP, Python, SQL, and ETL development.
- Built a **retail data pipeline** for sales trend analysis and forecasting using structured datasets.

PROJECTS

Automated Sales Data Validation DAG, *Apache Airflow, BigQuery, SQL, Python, Power Automate*

- Engineered a configurable **Apache Airflow data pipeline** to automate sales **data validation** across source and target systems using CTE-based SQL variance logic and **BigQuery** audit logging — cutting manual validation effort by ~70% and catching data discrepancies within minutes instead of hours.
- Implemented dynamic runtime configuration via **dag_run.conf**, enabling zero-code onboarding of new validation workflows and reducing pipeline setup time from days to under 30 minutes; integrated **real-time alerts** via Microsoft Teams using **Power Automate REST API**.

AI-Driven Data Quality Monitoring & Anomaly Detection, *BigQuery ML, SQL, Gemini 2.5 Pro*

- Built an **AI-driven anomaly detection** system using **BigQuery ML** Linear Regression with IQR-based adaptive thresholds — replacing hardcoded rules and improving anomaly detection accuracy to ~92% with a false positive rate under 5% across 6 months of backtested data.
- Integrated **Gemini 2.5 Pro** via **Vertex AI** for automated **root-cause analysis** on flagged anomalies, reducing mean investigation time from ~45 minutes to under 5 minutes and enabling **real-time data quality** monitoring at scale.

AI AutoHeal Agent – Automated Job Failure Remediation, *GCP Cloud Run, BigQuery, Airflow, Gemini 2.5 Pro, Python, SQL*

- Architected an autonomous **AI agent** on **GCP Cloud Run** that detects **Airflow DAG** failures from **BigQuery** and auto-remediates recoverable issues using **Gemini 2.5 Pro** classification — resolving ~75% of transient failures without human intervention and cutting on-call response time by ~60%.
- Implemented a **data quality** guardrail to block auto-remediation for integrity-related failures, preventing incorrect fixes; built end-to-end audit tracking with **NDJSON** logs loaded into **BigQuery**, achieving 100% job-level traceability across all remediation events.

EDUCATION

B.Tech (Artificial Intelligence), *G. H. Raisoni College of Engineering, Nagpur* 2020 – 2024 | Nagpur, India
CGPA: 8.83

CERTIFICATES

Google Cloud Certified Associate Data Practitioner

Discover Data Analysis Badge by Microsoft

PUBLICATIONS

Intelligent Diabetes Predicting Model for Diverse Ethnicities, *Springer Nature Singapore*

Transforming Operations: Integrating Computer Vision and Machine Learning for Automated Business Processes, *ICIPB Conference*

Deepfake Detection Using LSTM And ResNext, *IJCRT Conference*